

ASHFORD UNIVERSITY MEDICAL CENTER

Department of Pathology & Surgical Pathology

2200 Ridgewell Parkway, Pathology Building B, Hartford, CT 06103

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(fictional identifiers — sample only)

Surgical Pathology & Molecular Diagnostic Report

SPECIMEN ID: SAMPLE-0000000

A PATIENT DEMOGRAPHICS & SPECIMEN METADATA

PATIENT NAME

Arthur Pendelton (fictional)

PATIENT ID / MRN

SAMPLE-31109

DATE OF BIRTH

09/03/1959 (Age 66)

ORDERING PHYSICIAN

Dr. Marcus Vance, MD — Urologic Oncology

GENDER

Male

REPORTING PATHOLOGIST

Dr. Elizabeth Thorne, MD, FCAP

SPECIMEN TYPE

FFPE Core Biopsy (Formalin-Fixed Paraffin-Embedded)

COLLECTION DATE

August 2, 2026

PRIMARY SPECIMEN SOURCE

Bone Lesion Core Biopsy (Right Iliac Crest)

REPORT DATE

August 6, 2026

B CLINICAL DIAGNOSIS & HISTOPATHOLOGY SUMMARY

Metastatic Castration-Resistant Prostate Cancer (mCRPC)

Histologic Grade: Gleason Score 5+4 = 9 (Grade Group 5)

Clinical Staging: Stage IVB — T3a, N1, M1b (bone metastases predominant)

ECOG Performance Status: 1 — Restricted in strenuous activity

BIOMARKER	RESULT	METHOD
PSA	48.5 ng/mL	Immunoassay
AR-V7	Negative	RT-PCR
PSMA Expression	High — IHC 3+	IHC

C NEXT-GENERATION SEQUENCING — GENOMIC PROFILE

GENE	VARIANT / ALTERATION	ALLELE FREQ.	CLASSIFICATION	CLINICAL SIGNIFICANCE
ATM	c.1564_1565del (p.Glu522fs)	39.4% (Somatic)	Pathogenic	DNA repair defect / PARP inhibitor candidate
PTEN	Homozygous Deletion (Exons 1–9)	N/A	Pathogenic	Akt/PI3K pathway activation

Gene	Variant / Alteration	Allele Freq.	Classification	Clinical Significance
AR	Amplification (Copy Number: 12)	N/A	Pathogenic	Anti-androgen resistance
Tumor Mutational Burden		Microsatellite Status		
4.8 Mut/Mb		Stable (MSS)		

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Treatment History

- JAN 2024 – JUN 2026 — PROGRESSION

1st-Line Therapy

Androgen deprivation therapy (Leuprolide) + Enzalutamide; progression noted June 2026
- THROUGH MAY 2026 — COMPLETED

2nd-Line Therapy

Docetaxel chemotherapy, 6 cycles

E

Current Laboratory & Organ Function Panels

as of Aug 6, 2026

Parameter	Result	Reference Range	Status
Platelet Count	88 ×10 ³ /μL	150 – 450	Grade 2 Thrombocytopenia
Hemoglobin	9.1 g/dL	13.8 – 17.2	Mild Anemia
Absolute Neutrophil Count (ANC)	1.6 ×10 ³ /μL	1.5 – 8.0	Low Normal
eGFR (Kidney Function)	82 mL/min/1.73m ²	> 60	Normal Renal Function
Serum Bilirubin	0.8 mg/dL	0.2 – 1.2	Normal
CNS Status	No brain metastases detected	—	Negative for CNS Lesions

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Pathologist Impression & Recommendation

Impression

Tumor exhibits biallelic inactivation of PTEN alongside a pathogenic ATM frameshift mutation. Bone marrow infiltration contributes to borderline platelet depression (88 K/μL).

Recommendation

Patient qualifies for targeted PARP inhibitor protocol evaluation or AKT inhibitor combination trials, pending platelet recovery or protocol waiver approval.

Dr. Elizabeth Thorne, MD, FCAP

Director of Surgical Pathology

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